



PSDL LAB

Information technology (Savitribai Phule Pune University)



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PSDL Practicals

1. Study of Embedded C programming language (Overview, syntax, One simple program like addition of two numbers).

```
#include<PIC18F458.h>
#include<stdio.h>
void main(){
    int a,b,c;
    a=100;
    b=100;
    c=a+b;
    TRISB=0;
    PORTC=c;
}
```

2. Write an Embedded C program to add array of n numbers.

```
#include<PIC18F458.h>
#include<stdio.h>
void main(){
    int i,sum;
    int n[]={11,12,13,14,15};
    sum=0;
    for(i=0;i<=4;i++){
        sum=sum+n[i];
    }
    TRISB=0;
    PORTB=sum;
}
```

3. Write an Embedded C program to transfer elements from one location to another for following:

- i) Internal to internal memory transfer
- ii) Internal to external memory transfer.

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <PIC18F458.h>
#pragma config OSC=HS
#define SIZE 10
unsigned char data[SIZE];
unsigned char external_mem[SIZE];
void transfer_internal_to_internal(unsigned char *src,unsigned char *dest,int size)
{
    memcpy(dest, src, size);
}
void transfer_internal_to_external(unsigned char *src,unsigned char *dest,int size)
{
    memcpy(dest, src, size);
}
void main(void)
{

```

```

for( int i = 0; i<SIZE; i++)
{
data[i]=i;
}
unsigned char internal_data[SIZE];
transfer_internal_to_internal(data,internal_data,SIZE );
transfer_internal_to_external(data,external_mem,SIZE );
printf("Data in internal Memory : \n");
for (int i =0; i<SIZE; i++ ){
printf("%d",internal_data[i]);
}
printf("\n");
printf("Data in external Memory : \n");
for (int i =0; i<SIZE; i++ ){
printf("%d",external_mem[i]);
}
printf("\n");
while(1);
}

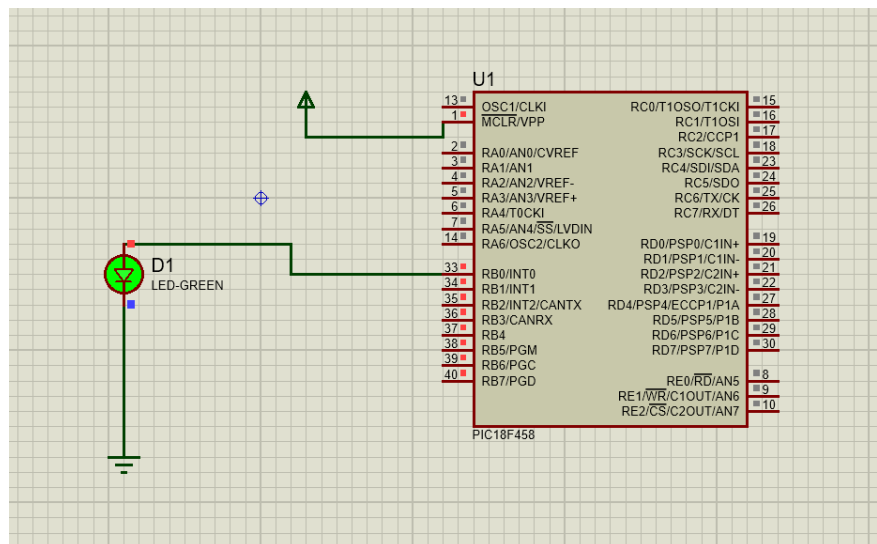
```

4. Write an Embedded C program to interface PIC 18FXXX with LED & blinking it using specified delay.

```

#include <PIC18F458.h>
// Define the delay in milliseconds
void delay();
void main(void) {
// Set the TRISB register to output mode for the LED
TRISB = 0x00;
// Loop forever
// Turn the LED on
while(1)
{
PORTB = 0xFF;
delay();
PORTB = 0x00;
delay();
}
}
void delay()
{
for(int
i=0;i<=10000;i++)
{
}
}

```



5. Write an Embedded C program for LCD interfacing with PIC 18FXXX.

```

#include <PIC18F458.h>
#define ldata PORTB
#define rs PORTCbits.RC1
#define en PORTCbits.RC0

```

```

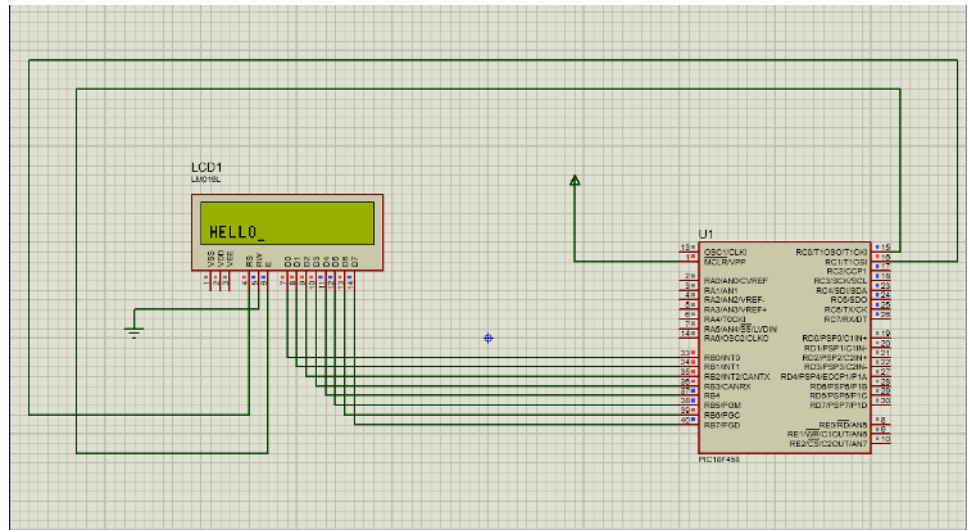
void delay();
void lcdcmd(unsigned char);
void lcdval(unsigned char);
void main()
{
    unsigned char xc;
    TRISB=0;
    TRISC=0;
    lcdcmd(0x38);
    lcdcmd(0x0E);
    lcdcmd(0x01);
    lcdcmd(0x06);
    lcdcmd(0xC0);
    lcdval('H');
    lcdval('E');
    lcdval('L');
    lcdval('L');
    lcdval('O');
    while(1);
}

void lcdcmd(unsigned char y)
{
    rs=0;
    ldata=y;
    en=1;
    delay();
    en=0;
    delay();
}

void lcdval(unsigned char y)
{
    rs=1;
    ldata=y;
    en=1;
    delay();
    en=0;
    delay();
}

void delay()
{
    for(int i=0;i<=2000;i++)
    {}
}

```



6. Write an Embedded C program for Generating PWM signal for DC motor.

```

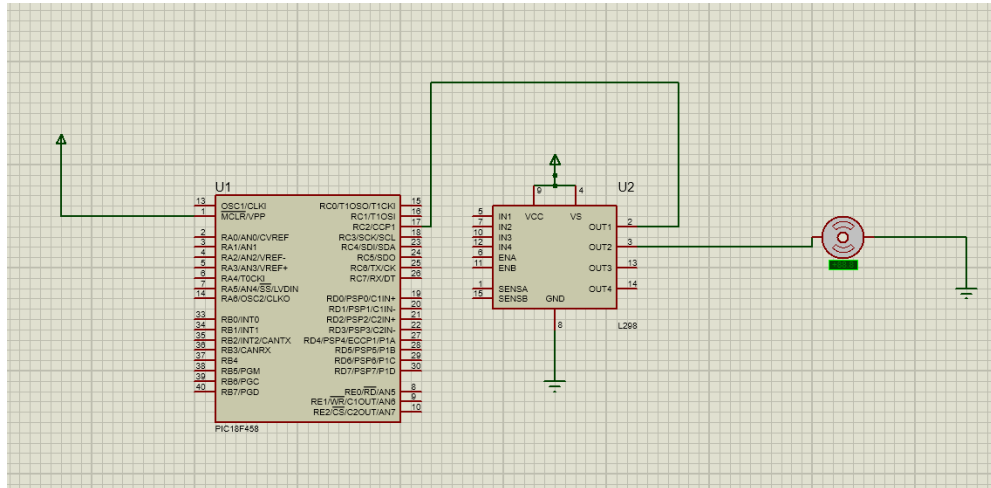
#include<pic18f458.h>
void main(void)
{
    TRISCbits.RC2 = 0;
    CCP1CONbits.CCP1M = 0b1100;
    TMR2ON = 0;
    while(1)

```

```

{
    PR2      = 74;
    CCPR1L   = 18;
    TMR2ON   = 1;
}
}

```



7. Write an Embedded C program for PC to PC serial communication using UART.

```

#include <PIC18F458.h>
void main()
{
    unsigned char a[]="WELCOME";
    int i;
    TRISB=0;
    TRISCbits.RC6=0;
    TRISCbits.RC7=1;
    SPBRG=0X4D;
    RCSTAbits.SPEN = 1;
    TXSTAbits.TXEN = 1;

    for(i=0;i<=6;i++)
    {

```

```

        while(PIR1bits.TXIF==0);
        TXREG=a[i];
        PIR1bits.TXIF=0;
    }

    while(1);
}

```

